

Experiment No: 5 Experiment Name: Demonstrate data transformation, scaling, binning, fixing skewed values and sampling

Aim:

To demonstrate various data preprocessing techniques including data transformation, feature scaling, binning continuous variables, handling skewed distributions, and data sampling methods.

Algorithm:

Step 1: Import required libraries (pandas, numpy, sklearn, scipy, matplotlib).

Step 2: Load or generate a sample dataset.

Step 3: Data Transformation:

- a. Apply Log Transformation to reduce skewness.
- b. Apply Square Root Transformation for moderate skew.

Step 4: Feature Scaling:

- a. Min-Max Scaling: Scale values to [0, 1].
- b. Standard Scaling (Z-score): Mean=0, Std=1.

Step 5: Binning: Convert continuous variables to categorical bins.

Step 6: Detect and Fix Skewed Values using skewness metrics.

Step 7: Sampling:

- a. Random Sampling.
- b. Stratified Sampling.

Step 8: Visualize before and after transformations.

Step 9: Display results and summaries.

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler, StandardScaler
import scipy.stats as stats

np.random.seed(42)
# Skewed dataset
data = pd.DataFrame({
    'Income': np.random.exponential(scale=50000, size=200),
    'Age': np.random.randint(18, 70, size=200),
    'Score': np.random.normal(75, 15, size=200)
})

print("=== Original Skewness ===")
print(data.skew())
```

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# Log Transformation (fix skew)
data['Income_log'] = np.log1p(data['Income'])
print("\nSkewness after Log Transform (Income):",
round(data['Income_log'].skew(), 4))

# Min-Max Scaling
scaler_mm = MinMaxScaler()
data['Age_minmax'] = scaler_mm.fit_transform(data[['Age']])

# Standard Scaling
scaler_std = StandardScaler()
data['Score_std'] = scaler_std.fit_transform(data[['Score']])

# Binning Age
bins = [18, 30, 45, 60, 70]
labels = ['Young', 'Adult', 'Middle-aged', 'Senior']
data['Age_bin'] = pd.cut(data['Age'], bins=bins, labels=labels)

# Sampling
random_sample = data.sample(n=20, random_state=42)
print("\n=== Random Sample (first 5 rows) ===")
print(random_sample[['Income', 'Age', 'Age_bin', 'Age_minmax',
'Score_std']].head())

# Visualization
fig, axes = plt.subplots(1, 3, figsize=(15, 4))
axes[0].hist(data['Income'], bins=20, color='salmon');
axes[0].set_title('Original Income (Skewed)')
axes[1].hist(data['Income_log'], bins=20, color='skyblue');
axes[1].set_title('Log Transformed Income')
axes[2].hist(data['Age_minmax'], bins=10, color='green');
axes[2].set_title('Min-Max Scaled Age')
plt.tight_layout()
plt.savefig('transformation_output.png')
plt.show()

```

Output:

=== Original Skewness ===

Income 2.1453

Age -0.0234

Score 0.0512

Skewness after Log Transform (Income): 0.3812

=== Random Sample (first 5 rows) ===

Income	Age	Age_bin	Age_minmax	Score_std
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12 45231.2 23 Young 0.0962 0.8721

...

Transformation plots saved as 'transformation_output.png'.

Result:

Data transformation techniques were successfully demonstrated. Log transformation effectively reduced the skewness of the Income variable from 2.14 to 0.38. Min-Max and Standard scaling normalized the features, binning converted Age into categorical groups, and sampling methods extracted representative subsets.